

NT Starter Pack

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Theorem 0.1: Fermat's Little Theorem

If p is a prime and a is an integer with $p \nmid a$, then

$$a^{p-1} \equiv 1 \pmod{p}.$$

Equivalently, for any integer a ,

$$a^p \equiv a \pmod{p}.$$

Theorem 0.2: Euler's Theorem

If n is a positive integer and a is an integer with $\gcd(a, n) = 1$, then

$$a^{\phi(n)} \equiv 1 \pmod{n},$$

where $\phi(n)$ is Euler's totient function, the number of integers in $\{1, 2, \dots, n\}$ coprime to n .

Theorem 0.3: Wilson's Theorem

A positive integer $p > 1$ is prime if and only if

$$(p-1)! \equiv -1 \pmod{p}.$$

Theorem 0.4: Fermat's Christmas Theorem

An odd prime p can be expressed as the sum of two squares,

$$p = a^2 + b^2, \quad a, b \in \mathbb{Z},$$

if and only if

$$p \equiv 1 \pmod{4}.$$

Theorem 0.5: Chinese Remainder Theorem

Let $n_1, n_2, \dots, n_k$ be pairwise coprime positive integers. Then for any integers $a_1, a_2, \dots, a_k$, the system of congruences

$$x \equiv a_1 \pmod{n_1}, \quad x \equiv a_2 \pmod{n_2}, \quad \dots, \quad x \equiv a_k \pmod{n_k}$$

has a unique solution modulo $N = n_1 n_2 \cdots n_k$.

Theorem 0.6: Wolstenholme's Theorem

Let $p \geq 5$ be a prime. Prove that $(p-1)!(\frac{1}{1} + \frac{1}{2} + \cdots + \frac{1}{p-1})$ is divisible by p^2 .

Theorem 0.7: Lifting The Exponent Lemma

Let p be an odd prime, and let $a, b \in \mathbb{Z}$ such that:

1. $p \nmid a$ and $p \nmid b$
2. $p \mid a - b$

Then for all $n \in \mathbb{Z}^+$:

$$v_p(a^n - b^n) = v_p(a - b) + v_p(n)$$

Theorem 0.8: LTE for $a^n + b^n$, odd prime

Let p be an odd prime, and let $a, b \in \mathbb{Z}$ such that:

1. $p \nmid a$ and $p \nmid b$
2. $p \mid a + b$

Then for all **odd** $n \in \mathbb{Z}^+$:

$$v_p(a^n + b^n) = v_p(a + b) + v_p(n)$$

Theorem 0.9: LTE Exception Case: $p = 2$

Let $a, b \in \mathbb{Z}$ with $2 \mid a - b$ (i.e., a and b are both odd). Then for all n is even:

$$v_2(a^n - b^n) = v_2(a - b) + v_2(a + b) + v_2(n) - 1$$

If n is odd, then

$$v_2(a^n - b^n) = v_2(a - b)$$

If n is even and $4 \mid a - b$:

$$v_2(a^n - b^n) = v_2(a - b) + v_2(n)$$

Note: For $p = 2$ and $a^n + b^n$, LTE does *not* apply in general.

Notation: $v_p(x)$ denotes the p -adic valuation of x , i.e., the largest exponent k such that $p^k \mid x$.

Problem 1

A student firstly wrote $x = 3$ on the board. For each process, the student deletes the number x and replaces it with either $(2x + 4)$ or $(3x + 8)$ or $(x^2 + 5x)$. Is this possible to make the number $(20^{17} + 2016)$ on the board?

Problem 2

a, b, c are positive integer. Solve the equation: $2^{a!} + 2^{b!} = c^3$

Problem 3

Find, with proof, all positive integers n for which $2^n + 12^n + 2011^n$ is a perfect square.

Problem 4

Suppose that the set $\{1, 2, \dots, 1998\}$ has been partitioned into disjoint pairs $\{a_i, b_i\}$ ($1 \leq i \leq 999$) so that for all i , $|a_i - b_i|$ equals 1 or 6. Prove that the sum

$$|a_1 - b_1| + |a_2 - b_2| + \dots + |a_{999} - b_{999}|$$

ends in the digit 9.

Problem 5

Let $p = 6967$ be a prime. Find the remainder when

$$\left\lfloor \frac{1^p}{p} \right\rfloor + \left\lfloor \frac{2^p}{p} \right\rfloor + \left\lfloor \frac{3^p}{p} \right\rfloor + \dots + \left\lfloor \frac{6965^p}{p} \right\rfloor$$

is divided by p . Here $\lfloor \cdot \rfloor$ denotes the greatest integer function.

Problem 6

Find all triples of positive integers (a, b, p) with p prime and

$$a^p + b^p = p!.$$

Problem 7

A positive integer a is selected, and some positive integers are written on a board. G8 and NNT play the following game. On G8's turn, he must replace some integer n on the board with $n + a$, and on NNT's turn he must replace some even integer n on the board with $n/2$. G8 goes first and they alternate turns. If on his turn NNT has no valid moves, the game ends.

After analyzing the integers on the board, NNT realizes that, regardless of what moves G8 makes, he will be able to force the game to end eventually. Show that, in fact, for this value of a and these integers on the board, the game is guaranteed to end regardless of G8's or NNT's moves.

1 Buffer Problems

Problem 8

Find all triples of positive integers (x, y, z) that satisfy the equation

$$2020^x + 2^y = 2024^z.$$

Problem 9

Find all pairs of distinct primes (p, q) such that

$$q^p \mid p + p^q + p^{q^p}$$

Problem 10

A set X consisting of n positive integers is called *PEAK* if the following condition holds: For any two different subsets of X , say A and B , the number $s(A) - s(B)$ is not divisible by 2^n . (Here, for a set A , $s(A)$ denotes the sum of the elements of A) Given n , find the number of *PEAK* sets of size n , all of whose elements is strictly less than 2^n .